

Lab 3 Architecture Justification

Self-Service Coffee Kiosk System | Dhyan Rao | PES1UG24AM090 | Section B

Architecture Selection

We chose a Layered Architecture for the Self-Service Coffee Kiosk System.

Reasons for the Choice

1. Clear separation of responsibilities: the touchscreen interface, order processing, payment handling, menu and pricing data, and receipt hardware can evolve independently while keeping the kiosk workflow understandable.
2. Suitable for a compact kiosk: the system has a small, stable set of functions and hardware integrations, so the lower operational complexity of layers is preferable to the deployment and network overhead of microservices.

Security Advantage

The payment service can be isolated behind a controlled interface. Card details are handled only by the payment component, while the order manager receives a success or failure result rather than raw payment data. This supports least privilege and reduces exposure of sensitive information.

Performance Benefit

Most menu lookups and order calculations remain local to the kiosk, reducing network calls and response time for touch interactions. The order manager can also validate the order before invoking the payment and printer interfaces, avoiding unnecessary hardware operations.

Components and Interfaces

The diagram contains five components: Touchscreen UI, Order Manager, Payment Service, Menu and Pricing Store, and Receipt Printer Adapter. It shows provided and required interfaces for payment processing, menu queries, receipt generation, card-terminal communication, database access, and printer communication.